

Student Information Form using HTML, CSS and JavaScript

Experiment Report — Internal, External and Inline JavaScript

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Aim

To design a Student Information web form using HTML and CSS, and to demonstrate the use of internal, external, and inline JavaScript for form validation, DOM manipulation, and console-based debugging output.

Software / Tools Required

- Visual Studio Code
- Google Chrome (or any modern web browser)
- HTML5
- CSS3
- JavaScript

Program Code

The webpage uses three types of JavaScript: **internal JavaScript** (inside a `<script>` tag in the `<head>`, showing a welcome alert and console logs on page load), **inline JavaScript** (the `onclick` attribute on the "Display Details" button), and **external JavaScript** (`script.js`, handling form validation and displaying entered details).

index.html

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <title>Student Webpage</title>
    <link rel="stylesheet" href="style.css">

    <!-- Internal JavaScript -->
    <script>
        function welcomeMessage() {
            alert("Welcome to the Student Information Portal!");

            console.log("Website Loaded Successfully");
            console.info("Internal JavaScript Executed");
            console.warn("This is a warning message.");
            console.error("Sample Error Message.");

            const student = [
                {
                    Name: "Parth Denge",
                    PRN: "24070521018",
                    Course: "B.Tech CSE"
                }
            ];

            console.table(student);
```

```

    }
</script>

</head>

<body onload="welcomeMessage()">

    <div class="container">

        <h1>Student Information</h1>

        <form onsubmit="return validateForm()">

            <label>Name</label>
            <input type="text" id="name">

            <label>PRN Number</label>
            <input type="text" id="prn">

            <label>Email</label>
            <input type="email" id="email">

            <label>Course</label>
            <input type="text" id="course">

            <!-- Inline JavaScript -->
            <button type="button"
                onclick="displayStudent(); alert('Student Details Displayed Successfully!')">
                Display Details
            </button>

            <button type="submit">
                Submit
            </button>

        </form>

        <hr>

        <h2>Student Details</h2>

        <p><b>Name:</b> <span id="dname"></span></p>
        <p><b>PRN:</b> <span id="dprn"></span></p>
        <p><b>Email:</b> <span id="demail"></span></p>
        <p><b>Course:</b> <span id="dcourse"></span></p>

    </div>

    <!-- External JavaScript -->
    <script src="script.js"></script>

</body>

</html>

```

style.css

```

body{

    font-family: Arial, sans-serif;
    background:#f2f2f2;
}

.container{

    width:500px;
    margin:auto;

```

```

        margin-top:40px;
        background:white;
        padding:25px;
        border-radius:10px;
        box-shadow:0px 0px 10px gray;
    }

    h1{

        text-align:center;
        color:darkblue;
    }

    label{

        display:block;
        margin-top:15px;
    }

    input{

        width:100%;
        padding:8px;
        margin-top:5px;
    }

    button{

        margin-top:20px;
        padding:10px 20px;
        background:green;
        color:white;
        border:none;
        cursor:pointer;
    }

    button:hover{

        background:darkgreen;
    }

    hr{

        margin-top:25px;
    }

```

script.js

```

function validateForm()
{

    let name=document.getElementById("name").value;
    let prn=document.getElementById("prn").value;
    let email=document.getElementById("email").value;
    let course=document.getElementById("course").value;

    if(name=="")
    {
        alert("Enter Name");
        return false;
    }

    if(prn=="")
    {
        alert("Enter PRN");
        return false;
    }
}

```

```

    if(email=="")
    {
        alert("Enter Email");
        return false;
    }

    if(course=="")
    {
        alert("Enter Course");
        return false;
    }

    if(!email.includes("@"))
    {
        alert("Invalid Email");
        return false;
    }

    console.log("Form Validation Successful");

    alert("Form Submitted Successfully");

    return true;
}

function displayStudent()
{

    document.getElementById("dname").innerHTML=
    document.getElementById("name").value;

    document.getElementById("dprn").innerHTML=
    document.getElementById("prn").value;

    document.getElementById("demail").innerHTML=
    document.getElementById("email").value;

    document.getElementById("dcourse").innerHTML=
    document.getElementById("course").value;

    console.info("Student Details Displayed");

    console.log("Display Function Executed");

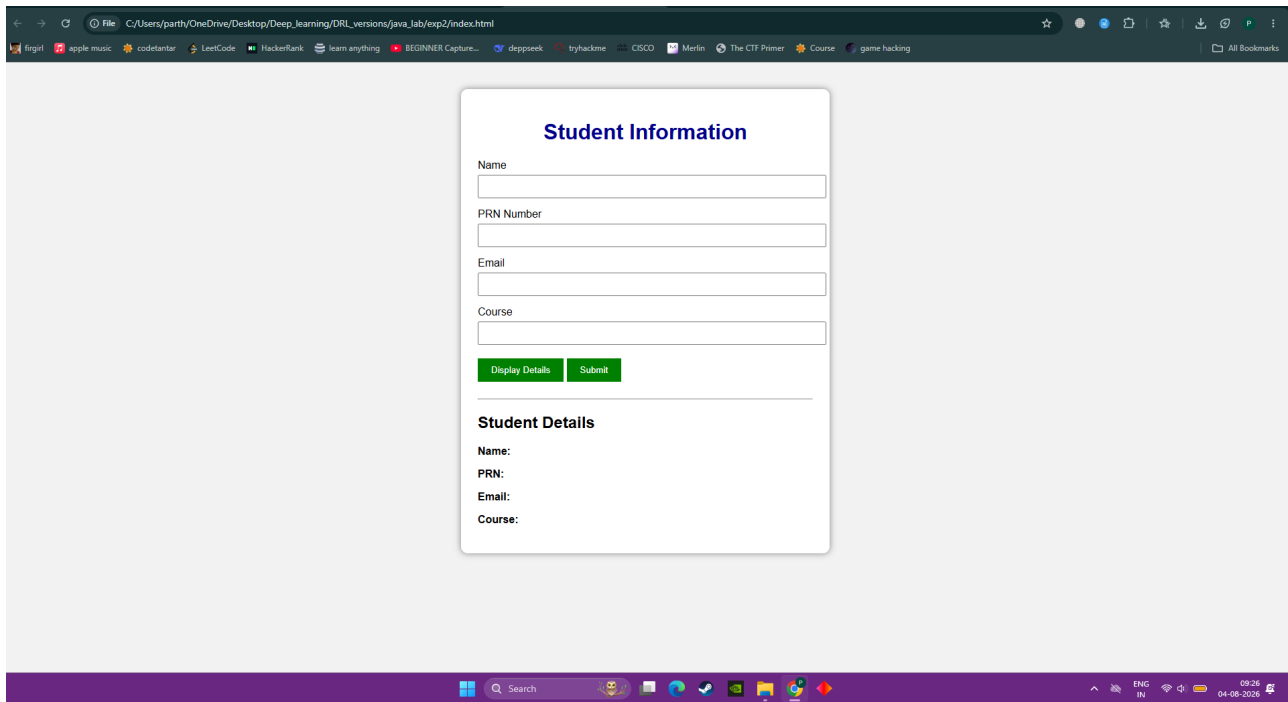
    console.warn("Always verify student information before submission.");

}

```

Output

The screenshot below shows the running Student Information form, loaded from the local file path shown in the browser's address bar.



Result / Conclusion

Result:

The Student Information Form was developed successfully using HTML, CSS, and JavaScript. The form correctly validates required fields and email format on submission, and displays the entered student details (Name, PRN, Email, Course) on the page when requested.

Conclusion:

This experiment demonstrates the difference between internal, external, and inline JavaScript, and how each can be used within the same webpage. It also covers form validation, DOM manipulation using `getElementById`, event handling (`onload`, `onclick`, `onsubmit`), and the use of various console methods (`log`, `info`, `warn`, `error`, `table`) for debugging.